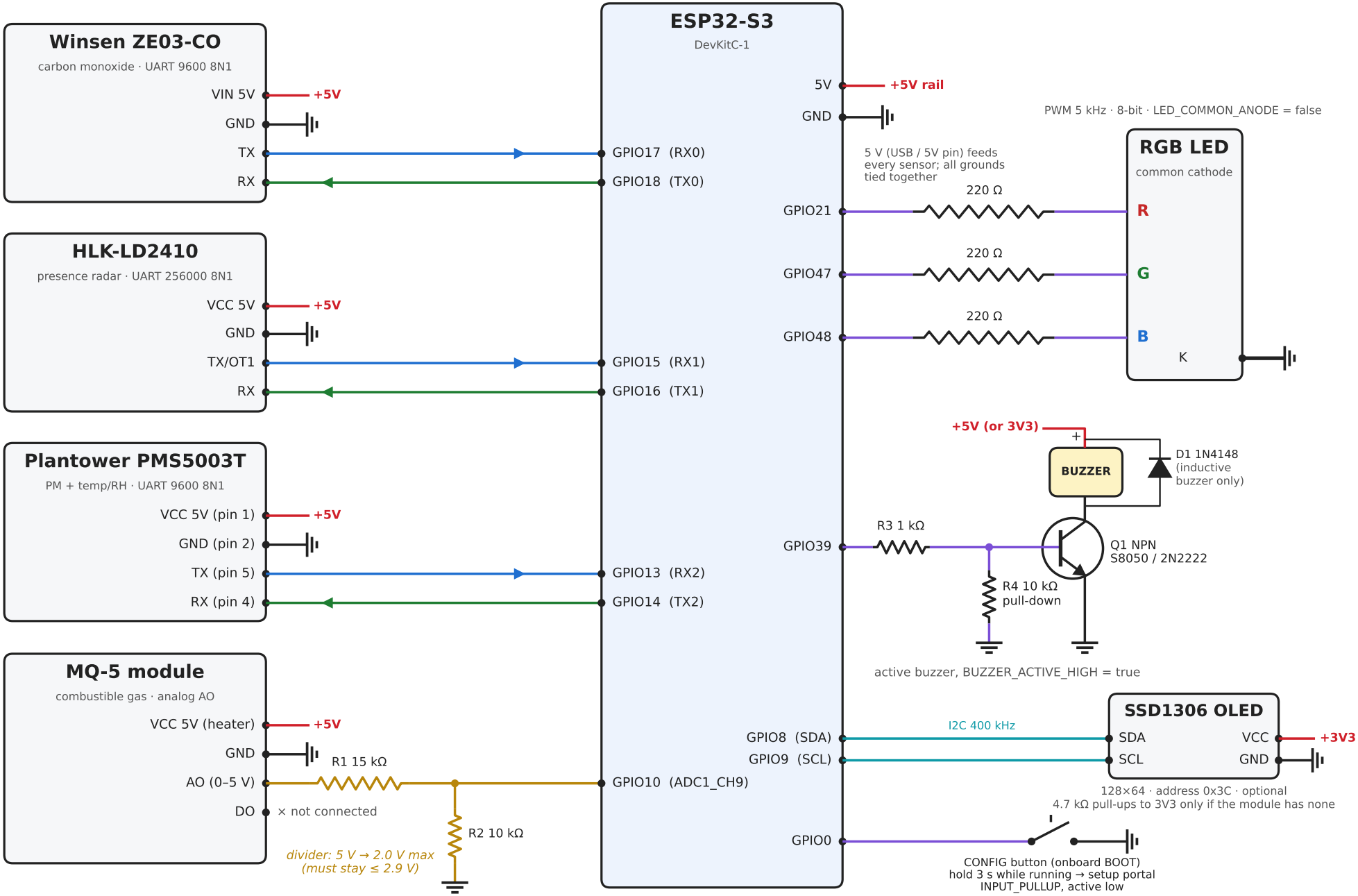
Canary air sensor — wiring diagram

ESP32-S3-DevKitC-1 · firmware 1.1.0 · pin map from air_sensor/HardwareConfig.h



Pin map and wiring notes

Function	ESP32-S3 pin	Connects to	Setting
ZE03-CO receive	GPIO17 (UART0 RX)	ZE03 TX	9600 8N1, CO_RX_PIN
ZE03-CO transmit	GPIO18 (UART0 TX)	ZE03 RX	CO_TX_PIN
LD2410 receive	GPIO15 (UART1 RX)	LD2410 TX	256000 8N1, PRESENCE_RX_PIN
LD2410 transmit	GPIO16 (UART1 TX)	LD2410 RX	PRESENCE_TX_PIN
PMS5003T receive	GPIO13 (UART2 RX)	PMS TX (pin 5)	9600 8N1, PARTICULATE_RX_PIN
PMS5003T transmit	GPIO14 (UART2 TX)	PMS RX (pin 4)	PARTICULATE_TX_PIN
MQ-5 analog	GPIO10 (ADC1_CH9)	MQ-5 AO via R1/R2 divider (×0.4)	12-bit, 11 dB
LED red	GPIO21	220 Ω → LED R anode	PWM 5 kHz, 8-bit
LED green	GPIO47	220 Ω → LED G anode	LED_GREEN_PIN
LED blue	GPIO48	220 Ω → LED B anode	LED_COMMON_ANODE = false
Buzzer driver	GPIO39	1 kΩ → NPN base	BUZZER_ACTIVE_HIGH = true
Config button	GPIO0	button to GND (onboard BOOT)	INPUT_PULLUP, hold 3 s
OLED data (optional)	GPIO8 (I2C SDA)	SSD1306 SDA	400 kHz, 0x3C, DISPLAY_SDA_PIN
OLED clock (optional)	GPIO9 (I2C SCL)	SSD1306 SCL	DISPLAY_SCL_PIN
Power	5V / 3V3 / GND	5V: all four sensors · 3V3: OLED	shared ground required

Notes

1. MQ-5 divider is mandatory. AO swings up to ~5 V; the ADC resolves only ~0–2.9 V at 11 dB and a 5 V input damages the pin. R1 = 15 kΩ (AO side), R2 = 10 kΩ (GND side) gives 2.0 V at full scale; swapped they give 3.0 V and clip. Default gas thresholds (warn 1200, danger 1760 counts) assume this ratio.
2. MQ-5 heater draws ~150 mA at 5 V. Feed it from the 5 V rail, never from a GPIO.
3. All sensor signals are 3.3 V TTL (ZE03, LD2410, PMS5003T run on 5 V but talk at 3 V levels) — wire them directly, no level shifter.
4. Cross TX/RX: each sensor's TX goes to the ESP RX pin listed above, and its RX to the ESP TX pin.
5. Build with “USB CDC On Boot: Enabled”. UART0 belongs to the ZE03-CO; the debug console is on the native USB port.
6. Confirm the particulate sensor is a PMS5003T, not a plain PMS5003 — the parser cannot tell them apart and would report particle counts as temperature.
7. Do not use GPIO35/36/37 for the LED on modules with Octal PSRAM (N8R8, N16R8, every WROOM-2). The compiled GPIO21/47/48 set works on every module. GPIO48 also drives the onboard RGB LED on DevKitC-1 v1.0.
8. GPIO39 has a weak pull-up at reset, so the buzzer can chirp at power-on. R4 (10 kΩ pull-down) suppresses it. Add D1 only if the buzzer is inductive (magnetic type).
9. GPIO0 is a boot strapping pin: holding it during reset enters ROM download mode, not the setup portal.
10. LD2410: B/C variants ship at 256000 baud; an older V1.03 LD2410 ships at 57600 — change PRESENCE_BAUD if no frames arrive. LD2410S/D are not compatible.
11. Never open Serial2 without explicit pins: its default GPIO19/20 are the USB data lines.
12. The SSD1306 is optional. Power it from 3V3, not 5V. Address 0x3C (0x3D with the module's jumper moved, DISPLAY_I2C_ADDRESS). With no display the firmware logs it and runs unchanged; set DISPLAY_ENABLED = false to skip the probe. The alarm never depends on the display.
13. Resistor values for R1–R4, the LED resistors and the optional I2C pull-ups are recommendations, not fixed by the firmware.